Devan Soliman

▼ SOLIMAN

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Education

STANFORD UNIVERSITY – BACHELORS OF SCIENCE IN ELECTRICAL ENGINEERING Dec. 2025

RELEVANT COURSEWORK: Electricity/Magnetism, Semiconductor Devices, Circuits, Digital System Design, VLSI, Hardware Architectures, Object-Oriented Programming, Embedded Systems, Operating Systems

Skills

RTL + MICROARCHITECTURE
Verilog/SystemVerilog, FSMs, pipelining, simulation, high-level synthesis (Cadence Stratus & Siemens Catapult)

SYNTHESIS + EDA
Synopsys Design Compiler, static timing analysis, Cadence JasperGold, Virtuoso Studio, Xcelium, SPICE, Vivado Design Suite, PPA optimization

SOFTWARE + FIRMWARE
Assembly, Bash, C/C++/C#/SystemC, Git, Java, Linux, Make, MATLAB, Python, Tcl

LAB + PROTOTYPING
additive manufacturing, Autodesk Inventor, composites, FPGAs, logic analyzers, oscilloscopes, signal generators, soldering

Projects

BRUISER – 3D-Printable Skateboards 2023 – present
Launched a series of high-performance skateboards and longboards designed to be fabricated on desktop 3D-printers and reinforced by a custom carbon fiber laminating process requiring minimal tooling. Currently iterating and field testing an open-source electrification mod and printable wheels. Backed by Stanford Moonshot Club.

ACTIONPI – the World's Smallest Open-Source Action Camera 2022 – present
Building video capturing software, OLED graphical user interface, hot-swappable Li-ion power supply, and high-durability enclosures compatible with various mounting systems and Pi Zero form-factor single-board computers.

MIPS PROCESSOR 2025
Implemented a 5-stage MIPS processor in Verilog. Developed the processor's datapath, control unit, arithmetic logic unit, and memory components, ensuring proper execution of MIPS instructions. Integrated pipeline stages for increased throughput and wrote control logic for hazard handling. Wrote testbenches to verify design.

RASTERIZER IC 2024
Synthesized a rasterizer integrated circuit that converts 3D models into raster images for display devices. Features up to 64x multisample anti-aliasing and optimizations to reduce triangle render time by up to 51%, dynamic power by 24%, and area by 41%. Fastest configuration could achieve over 71M triangles per second when simulated in Synopsys DC.

FPGA MUSIC PLAYER + VISUALIZER 2023
Programmed an AMD Zynq FPGA into a music player using Vivado. Player read notes and time values from a ROM and generated sine waves as audio output in real-time. Supported combining simultaneous notes into chords. Expanded functionality included track selection, fast-forward, rewind, and displaying waveforms via HDMI.

BARE METAL CONSOLE + WIRELESS CHAT 2022
Built a desktop terminal running bare-metal on a BCM2385 ARM CPU. Programmed memory management system, drivers for input and graphics, shell, and commands. Integrated reliable two-way wireless communication platform by writing ESP32 firmware, microcontroller driver, and chatroom application.

Experience

LAB ASSISTANT, LAB64 MAKERSPACE (Stanford University) Sept. 2025 – Dec. 2025

- Operated Stanford's electrical engineering makerspace to promote safe and efficient lab usage
- Trained students and researchers in using lab equipment, including power supplies, oscilloscopes, soldering stations, reflow ovens, machine tools, laser cutters, and 3D printers
- Supported users in troubleshooting hardware and prototyping challenges to encourage self-sufficiency and technical confidence

RESEARCH ASSISTANT, VIRTUAL HUMAN INTERACTION LAB (Stanford University) Sept. 2023 – June 2024

- Built interactive virtual/augmented reality experiments using Unity Engine with Stanford researchers
- Created an augmented reality exhibit for prehistoric swamp fauna with the California Academy of Sciences

DATA ANALYST INTERN, HUBBUB July 2022 – Sept. 2022

- Wrote software to monitor and visualize the prevalence of the monkeypox outbreak in the United States
- Leveraged Amazon Web Services and Microsoft Power Platform to process and store data